Pardeep Kumar

SCIENTIFIC COMPUTING RESEARCHER | NUMERICAL METHODS | ASPIRING EDUCATOR

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PROFILE

Scientific computing researcher with a submitted doctoral thesis from Delft University of Technology and more than ten years of experience in numerical modelling, simulation software, and computational problem-solving. My research at CWI Amsterdam concerns robust numerical methods for multiphase thermodynamics and CO2-rich transport.

I am now exploring a serious transition toward high-school education. Teaching brings together the parts of research I value most: asking careful questions, making difficult ideas understandable, and helping someone gain the confidence to reason independently. I would bring broad examples, patient explanation, and an honest learning mindset to mathematics, physics, and computing education.

TEACHING DIRECTION

I am preparing for a deliberate move toward high-school education. I want to help students approach mathematics and physics through careful questions, physical intuition, visual explanation, and small computational investigations.

- Interested in high-school mathematics, physics, computational thinking, and science outreach.

- Able to connect classroom ideas to real examples from fluids, thermodynamics, electromagnetics, aerospace, and engineering.

- Committed to treating questions and mistakes as useful parts of learning, while helping students learn how to test an argument.

EDUCATION

PhD in Mechanical Engineering **2022-2026**

Delft University of Technology | Research conducted at CWI Amsterdam

Thesis submitted; defence planned for January 2027. Research focus: numerical methods for transient multiphase flow of multicomponent mixtures, with industrial collaboration from Shell Projects & Technology.

MSc in Aerospace Engineering **2012-2014**

Indian Institute of Technology Bombay, India

Research focus: finite-volume time-domain methods for electromagnetic propagation and scattering.

DOCTORAL RESEARCH AND SCIENTIFIC CONTRIBUTION

- Reformulated constrained phase-equilibrium calculations in a more efficient thermodynamic variable space.
- Coupled phase-stability and equilibrium calculations to finite-volume solvers for CO2-rich tank and pipeline transport.
- Developed thermodynamically consistent temperature-evolution models for multiphase flow.
- Investigated exact and approximate Riemann problems for real-fluid mixtures.
- Created Julia and Python research software supported by systematic validation and reproducibility workflows.

TRANSFERABLE STRENGTHS FOR EDUCATION

Clear explanation Used to moving between mathematics, physical meaning, algorithms, plots, and plain-language interpretation.	Interdisciplinary perspective Examples drawn from aerospace, energy, fluids, electromagnetics, semiconductors, finance, and computing.
Evidence and verification A research habit of checking units, assumptions, limiting cases, numerical results, and alternative explanations.	Computational demonstrations Able to create small programs and visual models that let students investigate a mathematical or physical question.

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SCIENTIFIC COMPUTING, ENGINEERING EXPERIENCE, AND SELECTED OUTPUT

SCIENTIFIC COMPUTING STRENGTHS

Numerical methods Finite-volume and finite-difference methods, Riemann solvers, nonlinear equations, constrained optimisation, automatic differentiation, and numerical linear algebra.	Computational physics Multiphase flow, real-fluid thermodynamics, CFD, electromagnetics, wave propagation, and numerical relativity.
Scientific programming C++, Julia, Python, C#, F#, Java, MATLAB, CUDA, MPI, multithreading, and performance-oriented computation.	Research software practice Modular architecture, interoperability, testing, validation, version control, visualisation, and reproducible computation.

PROFESSIONAL EXPERIENCE

PhD Researcher in Scientific Computing — **Aug 2022-present**

Centrum Wiskunde & Informatica (CWI), Amsterdam

- Developed numerical formulations and software for multicomponent, multiphase thermodynamics and transient flow.
- Connected mathematical analysis, physical modelling, algorithm design, implementation, and validation in interdisciplinary research.

Software Engineer — **Feb-Aug 2022**

ASM International, Almere

- Enhanced and refactored a Java-based simulator for semiconductor thin-film deposition equipment.

Software Engineer — **Jun 2019-Dec 2021**

Shell, Amsterdam

- Migrated numerical workflows from Python to C++ and CUDA, reducing a representative runtime from about 30 minutes to 15 seconds.
- Developed concurrent infrastructure, interoperability layers, and visualisation for process and pipeline simulation software.

Earlier engineering roles — **Jul 2014-Apr 2019**

Aakraya Research, KLA-Tencor, Altair, and Fluidyn

- Worked on trading infrastructure, semiconductor simulation, geometry and mesh tools, engineering software architecture, and computational physics.
- Developed a parallel finite-volume electromagnetic solver, higher-order schemes, shared-memory infrastructure, and scientific testing tools.

SELECTED PEER-REVIEWED PUBLICATIONS

1. P. Kumar and P. I. Rosen Esquivel. A Reformulation of UVN-Flash for Multicomponent Two-Phase Systems with Application to CO2-Rich Mixture Transport in Pipelines. Computers & Fluids, 314, 107108, 2026.

2. P. Kumar and P. I. Rosen Esquivel. Solving the UVN-Flash Problem in TVN-Space. Fluid Phase Equilibria, 599, 114528, 2026.

3. P. Kumar, B. Sanderse, P. I. Rosen Esquivel, and R. A. W. M. Henkes. A New Temperature Evolution Equation That Enforces Thermodynamic Vapour-Liquid Equilibrium in Multiphase Flows. Computers & Fluids, 289, 106524, 2025.

WEB APPLICATIONS - SECONDARY INTEREST

I independently developed The Republic, Nazar India, Chess Duel, and WorkAtlas. The work demonstrates product thinking and self-directed learning, while remaining secondary to my scientific-computing and teaching direction. View web application experience

LANGUAGES

English: proficient | Hindi: proficient | Dutch: A2